

Standardized Mean Difference (SMD) vs. Standardized Mean Change (SMC)

Both the Standardized Mean Difference (SMD) and the Standardized Mean Change (SMC) express results in standardized units, allowing researchers to compare findings across studies. However, they apply to different research designs and interpret different forms of change.

1. Standardized Mean Difference (SMD)

The Standardized Mean Difference measures the difference between two **independent group means** in standardized units. It is used in between-subjects designs, such as comparing a treatment and control group.

Formula: $SMD = (M_1 - M_2) / SD_{pooled}$

Where:

M_1 = Mean of treatment group

M_2 = Mean of control group

$SD_{pooled} = \sqrt{(SD_1^2 + SD_2^2) / 2}$

Common estimators include Cohen's d, Hedges' g, and Glass's Δ .
SMD answers the question: "How different are the two groups?"

2. Standardized Mean Change (SMC)

The Standardized Mean Change measures the magnitude of change **within the same group** over time, such as pre-test vs. post-test results. It is used in within-subjects or repeated-measures designs.

Formula: $SMC = (M_{post} - M_{pre}) / SD_{pre}$

Alternatively, it can be standardized by the **SD of the change scores** or a pooled SD. Because pre- and post-scores are correlated, SMC often includes a correction for the pre-post correlation. Common estimators include variants described by Morris & DeShon

(2002), such as:

- Raw change standardized by pre-SD (dpre)
- Change standardized by the SD of change scores (dchange)
- Adjusted versions accounting for pre-post correlations

SMC answers the question: “How much did the group change over time?”

3. Conceptual Comparison

SMD (Standardized Mean Difference)

Between-subjects design

Compares means between two independent groups

Variance reflects between-group differences

Assumes groups are independent

Used for treatment vs. control comparisons

Typical estimators: Cohen’s d, Hedges’ g, Glass’s Δ

SMC (Standardized Mean Change)

Within-subjects design (repeated measures)

Compares pre- and post-means for the same group

Variance reflects within-person changes

Accounts for correlation between time points

Used for pre-post interventions or longitudinal change

Typical estimators: Morris & DeShon d_{av}, d_{change}

4. Example Calculations

Example 1 — SMD (Between Groups):

Treatment mean = 80, Control mean = 70, SD_{pooled} = 10

$$\text{SMD} = (80 - 70) / 30 = 0.33$$

Example 2 — SMC (Within Group):

Pre = 70, Post = 80, SD_{change} = 15

$$\text{SMC} = (80 - 70) / 15 = 0.66$$

Although numerically similar, SMD and SMC describe fundamentally different effects: SMD measures group differences, while SMC measures within-group change.

5. Selected References

Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum Associates.

Morris, S. B., & DeShon, R. P. (2002). *Combining effect size estimates in meta-analysis with repeated measures and independent-groups designs.* Psychological Methods, 7(1), 105–125.

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to Meta-Analysis.* Chichester, UK: Wiley.